MOMEN IBRAHIM FAWZY

Senior AI Engineer · Generative AI · LLMs & RAG · AI Agents · Arabic NLP

Riyadh, Saudi Arabia | momenibrahimfawzy982000@gmail.com
linkedin.com/in/momen-ibrahim-fawzy | github.com/Momen-Ibrahim-Fawzy
Google Scholar | ORCID 0009-0001-7287-0705 | momen-ibrahim-fawzy.github.io

Summary

Senior AI Engineer with 4 years building **production Generative AI** for Saudi government and enterprise clients, including the General Court of Audit, GAFT, CST, Al Rajhi Bank and Al Rajhi Capital. Builds **RAG platforms, tool-calling AI agents, bilingual Arabic/English voice assistants** and **ML decision systems**, end to end from requirements and architecture to Dockerized delivery, on-site in Riyadh since 2024. NLP researcher with **8 papers**, **three 1st places** and **three 2nd places** in international shared tasks (PAN@CLEF 2023 and 2026, ArabicNLP 2026), plus a paper at **IEEE ICDM 2026**. M.Sc. candidate, Alexandria University.

Professional Experience

EJADA Systems
Riyadh, KSA (on-site, Feb 2024 – present) · Alexandria, Egypt

Senior AI Engineer, Enterprise GenAI & AI Products — Oct 2025 – Present

- **General Court of Audit (GCA)**: lead the GenAI and AI-agent platform; built configurable LLM pipelines that filter, semantically classify and extract the legal hierarchy (regulation, article, paragraph) from audit-feedback records, with resumable batching and LLM fallback.
- **Al Rajhi Capital, Capital AI**: context-aware "copilot of agents" that orchestrates the RAG knowledge base and the robo-advisory engine (signals, forecasts, portfolio optimizer) through LLM tool calling, with streamed charts and human-in-the-loop trade proposals.
- **General Authority for Foreign Trade (GAFT), Arif** (co-developed): bilingual trade-analytics assistant using a LangChain SQL agent, query disambiguation, interactive Plotly charts, Excel export and SSE streaming.
- **Health-check product**: contactless health assessment that derives 30+ vital metrics from a 30-second webcam scan (rPPG with POS/CHROM, MediaPipe), with face-identity verification, an LLM health assistant and lab-report reading. **Message routing**: Arabic/English message classification and routing (topic, priority, urgency, sentiment).
- Set technical direction with stakeholders and represent the AI team on-site; contributed to the Saudi Water Authority (SWA)'s first AI-enablement initiative.

AI Engineer — Jul 2023 – Sep 2025

- **Voice assistant**, EJADA's bilingual voice and text assistant: microservices for speech-to-text on NVIDIA Triton, male/female Arabic TTS with diacritization, streaming LLM chat, hierarchical document parsing with Milvus retrieval, and conversation analytics. Delivered client editions.
- **Email intelligence**: 19 email-intelligence microservices behind an API gateway (tone, topics, semantic search, summaries, action items, reply suggestions) on a quantized LLM server; communication intelligence for the Communications, Space & Technology Commission (CST).
- **ServiceNow AI agent**: natural-language-to-SQL with conversation memory, query disambiguation, persona-aware prompts and automatic charts (LIDA).
- **Al Rajhi Bank, VCRM project (2023)**: AI and NLP capabilities for the bank's VCRM initiative.
- RAG pipelines (LangChain, LlamaIndex, Milvus) with hybrid search, and FastAPI inference microservices (sentiment, classification, summarization, financial documents).

AI Engineer (part-time) & AI Engineer Intern — Aug 2022 – Jul 2023

- Built NLP pipelines (classification, sentiment, NER) and enterprise chatbot components while completing the B.Sc.

City of Scientific Research & Technological Applications (SRTA-City)
Alexandria, Egypt

AI-Driven 3D Visualization Intern — May 2022 – Aug 2022

- Captured and processed 3D scan data (laser, photogrammetry, motion capture) for object-recognition models.

Research & Publications

- **ALEXIS at DialectSentEval Subtask 2: Retrieval-Augmented, Preference-Tuned Generation for Arabic Sentiment Swap**
  M. Ibrahim, M. E. Hassib, A. M. Salem, M. Torki. *ArabicNLP 2026 Shared Tasks*, ACL Anthology. **1st of 15**; Sentiment Style Accuracy 0.958, BLEU 56.9.
- **ALEXIS at DialectSentEval Subtask 1: Adversarial Training and Iterative Self-Training for Zero-Shot Dialect Sentiment Analysis**

M. Ibrahim, M. E. Hassib, A. M. Salem, M. Torki. *ArabicNLP 2026 Shared Tasks*, ACL Anthology. **1st of 12**; Macro-F1 0.867, +4.6 over the runner-up.

- **ALEXIS at AlexandriaX-2026 Subtask 2: Capacity Discovery and Calibrated Preference Optimization for Cross-Dialect Arabic MT**
  M. Ibrahim, M. E. Hassib, A. M. Salem, M. Torki. *ArabicNLP 2026 Shared Tasks*, ACL Anthology. **2nd of 4 teams**; NLLB-200 + LoRA + DPO, spBLEU 25.84.
- **Target-Aware Validation under Population Shift: Diagnosing Selection Bias in Small-Cohort Suicide Risk Detection**
  MHSM Workshop at IEEE ICDM 2026, Shenyang, China (accepted, to appear).
- **Post-Hoc Synonym Watermarking with Dual-Key Type-Level Detection**
  M. Ibrahim, N. El-Makky, M. Torki. *CLEF 2026*, CEUR-WS 4283. **2nd of 5 teams**, PAN Text Watermarking (TWF 0.775). Code.
- **Content-Agnostic Contrastive Stylometry for Multi-Author Writing Style Analysis**
  M. Ibrahim, N. El-Makky, M. Torki. *CLEF 2026*, CEUR-WS 4283. **2nd of 8 teams**, PAN Multi-Author Writing Style Analysis; F1 1.000 / 0.741 / 0.773. Code.
- **Disagreement-Aware Ensemble for Robust AI Text Detection**
  M. Ibrahim, N. El-Makky, M. Torki. *CLEF 2026*, CEUR-WS 4283. PAN Voight-Kampff Generative AI Detection. Code.
- **Enhancing Authorship Verification using Sentence-Transformers**
  M. Ibrahim, A. Akram, M. Radwan, R. Ayman, M. Abd-El-Hameed, N. El-Makky, M. Torki. *CLEF 2023*, CEUR-WS 3497. **1st place**, PAN 2023 Authorship Verification (62.3%). Code.

Graduate Course Projects

- **Arabic ChatGPT-text detection** (NLP): translated HC3 into an 86K-sample Arabic corpus; fine-tuned Arabic BERT detectors (CAMeLBERT, GigaBERT-v4) reached F1 0.995 in-domain, with out-of-domain tests and six adversarial-attack types (homoglyphs, phonetic and keyboard errors, segmentation).
- **Multi-author style-change detection** (NLP with DL, PAN 2024 data): fine-tuned sentence transformers (F1 0.989 / 0.880 / 0.839) outperformed GPT-3.5, Llama-3-70B and Mixtral on medium and hard documents.
- **Dental X-ray segmentation** (Image Segmentation, DENTEX data): U-Net and nnSAM multi-class tooth instance segmentation on panoramic X-rays, Dice 0.924 (quadrant) and 0.756 (tooth enumeration). **Image deblurring** (Image Processing): generalization study of FFTformer and SAM-Deblur.

Technical Skills

GenAI & LLMs	RAG, tool-calling agents, LLM guardrails, LangChain, LlamaIndex, llama.cpp, quantized LLMs, LoRA, DPO
Retrieval / Search	Milvus, Elasticsearch, semantic & hybrid search, BGE-M3 / sentence embeddings, MongoDB Vector Search
NLP & Speech	Arabic & dialectal NLP (MARBERT, CAMeLBERT, AraBERT), MT (NLLB), text-to-SQL, AI-text detection, stylometry, watermarking · STT (NVIDIA Triton), Arabic TTS
ML & Vision	PyTorch, Transformers, scikit-learn, LightGBM · rPPG, MediaPipe, U-Net, nnU-Net, SAM, YOLO, OCR
Engineering	Python, SQL, TypeScript · FastAPI, Flask, React, SSE streaming, microservices · Docker Compose, Kubernetes, PostgreSQL, MongoDB, Redis, Prometheus/Grafana

Education

M.Sc. Computer & Systems Engineering, Alexandria University 2023 – 2027 (expected)
Thesis: *Robust Semantic and Sentiment Preservation in Dialectal Arabic Machine Translation*. Supervisors: Prof. Nagwa El-Makky, Prof. Marwan Torki.

B.Sc. Computer & Systems Engineering, Alexandria University, GPA 3.68 / 4.0 2018 – 2023

Certifications, Memberships, Talks & Languages

- **IEEE Graduate Student Member** | **IEEE Young Professionals**
- Natural Language Processing, **Carnegie Mellon University** School of Computer Science (Emeritus) | **IBM** AI Workshop, Level 2
- MongoDB GenAI Modernization (Riyadh) | Machine Learning (Stanford) | Deep Learning (DeepLearning.AI) | **Languages:** Arabic (native), English (fluent), Italian (basic)
- Speaker at **CLEF 2026** (two PAN papers) | Author at **IEEE ICDM 2026** and **ArabicNLP 2026**